

# High-Speed Link Design

Lab Report 04

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## Slicer Characterization

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Cadence Virtuoso Simulation Analysis

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# 1 Introduction

High-speed serial links require a receiver to convert a small differential input signal into a reliable digital decision. The slicer is a key receiver block that performs this decision by comparing the differential input and regenerating the result into complementary logic-level outputs. For high-speed and low-power applications, dynamic slicer architectures are particularly useful because their operation is controlled by the clock, allowing the comparison and regeneration process to take place during a defined portion of each clock cycle.

This lab focuses on the operation, simulation, and characterization of a dynamic slicer. The circuit consists of a dynamic comparison stage followed by a regenerative stage and an SR latch that stores the final decision. The operation of the complete circuit is investigated through transient simulations, with particular emphasis on the delay and power consumption of the slicer. The effect of the differential input magnitude on the decision delay is also studied.

In addition to the basic transient characterization, the slicer is evaluated under an overdrive recovery condition. In this test, a large differential input is first applied for several cycles before a small opposite input is introduced. This allows the minimum input magnitude that can be correctly resolved after a large overdrive to be investigated. Finally, the offset voltage of the slicer is evaluated using a dedicated measurement setup and Verilog-A models. These characterizations provide a more complete understanding of the practical limitations of the dynamic slicer.

The lab is therefore divided into three main parts: basic dynamic comparator simulation and characterization, overdrive recovery testing, and offset-voltage simulation. The results from these experiments provide the necessary foundation for evaluating the performance of the slicer in terms of speed, power consumption, sensitivity, and offset voltage.

## 2 SR Latch Implementation and Verification

The dynamic slicer is followed by an SR latch to store the comparison result and provide complementary logic outputs. In this lab, a cross-coupled NOR-based SR latch was implemented and verified before integrating it with the dynamic comparator. The latch has two inputs, Set ( $S$ ) and Reset ( $R$ ), and two complementary outputs,  $Q$  and  $QB$ .

### 2.1 NOR-Based SR Latch

The implemented latch consists of two cross-coupled 2-input NOR gates. The output of the upper NOR gate is connected to  $Q$ , while the output of the lower NOR gate is connected to  $QB$ . Each output is fed back to one input of the opposite NOR gate, creating the required regenerative feedback.

The Boolean equations describing the latch are

$$Q = \overline{R + QB} \quad (1)$$

$$QB = \overline{S + Q} \quad (2)$$

where  $S$  and  $R$  are the set and reset inputs, respectively. Because NOR gates are active-high in this configuration, applying a logic high to  $S$  sets the latch, while applying a logic high to  $R$  resets it. When both inputs are low, the feedback maintains the previously stored state.

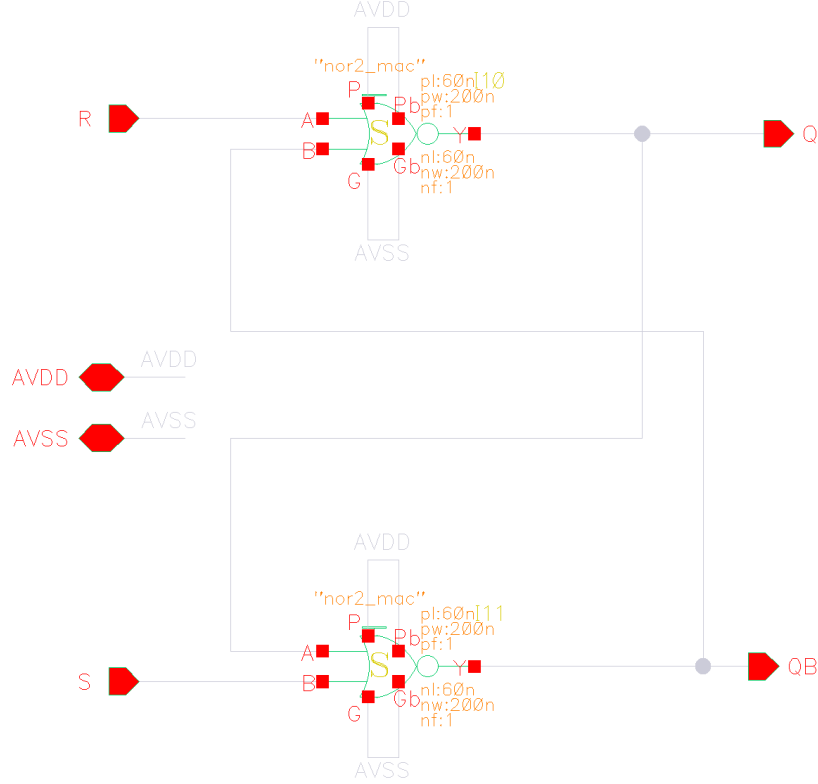


Figure 1: Implemented NOR-based SR latch.

The operation of the latch is summarized in Table 1.

| Table 1: Truth table of the NOR-based SR latch. |     |                |                              |
|-------------------------------------------------|-----|----------------|------------------------------|
| $S$                                             | $R$ | $Q$            | $QB$                         |
| 0                                               | 0   | Previous state | Complement of previous state |
| 1                                               | 0   | 1              | 0                            |
| 0                                               | 1   | 0              | 1                            |
| 1                                               | 1   | 0              | 0                            |

For  $S = 1$  and  $R = 0$ , the lower NOR gate forces  $QB$  to zero. Consequently, the upper NOR gate produces  $Q = 1$ , placing the latch in the set state. Similarly, for  $S = 0$  and  $R = 1$ , the upper NOR gate forces  $Q$  to zero and the latch enters the reset state.

When both inputs are zero, neither NOR gate is being directly forced by an external input. The cross-coupled feedback therefore maintains the existing state:

$$(S, R) = (0, 0) \Rightarrow Q(t) = Q(t^-) \quad (3)$$

The condition  $S = R = 1$  is the forbidden condition for a NOR-based SR latch because both outputs are forced low, violating the complementary relationship between  $Q$  and  $QB$ . Therefore, this condition should be avoided during normal operation.

### 2.1.1 Transient Verification

The functionality of the implemented latch was verified using transient simulation. Independent digital input signals were applied to  $S$  and  $R$ , and the resulting waveforms of  $Q$  and  $QB$  were observed.

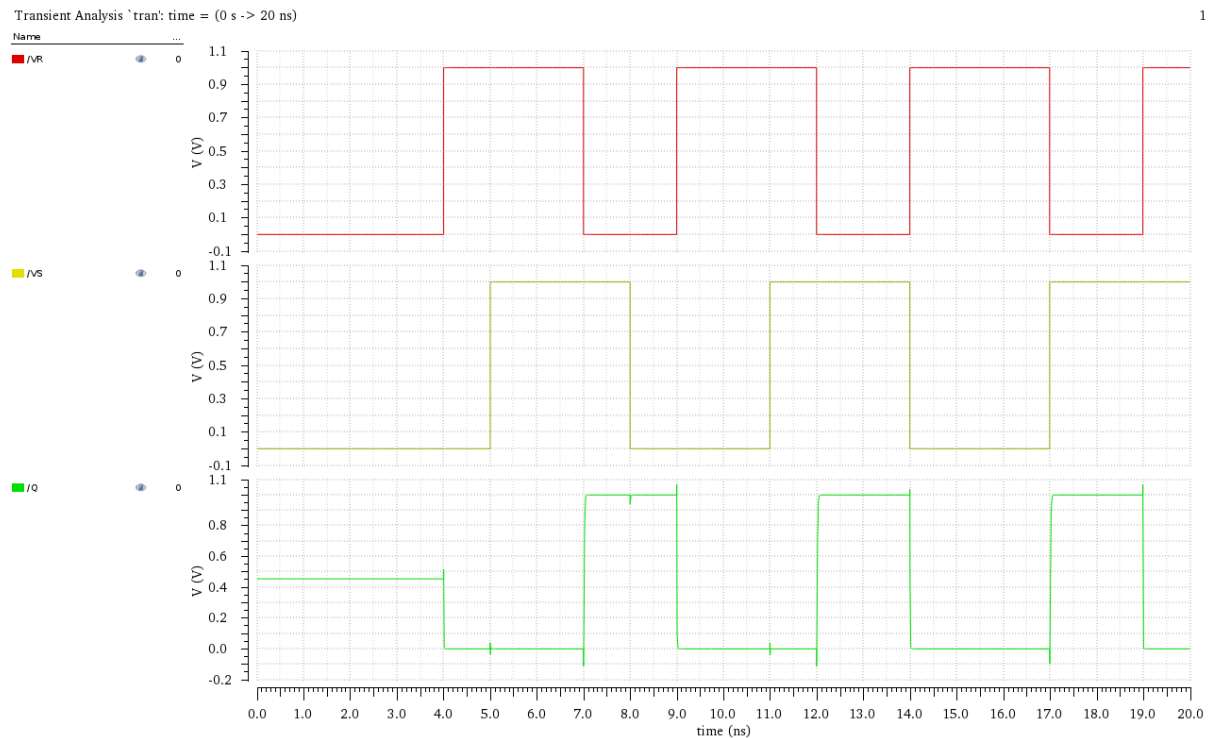


Figure 2: Transient simulation used to verify the SR-latch operation.

The simulated outputs follow the expected SR-latch behavior. When the reset input  $R$  is asserted, the output  $Q$  is driven low. When the reset condition is removed and the set input  $S$  is asserted,  $Q$  transitions to the high state. The complementary output  $QB$  exhibits the opposite logic state. When both control inputs return to zero, the cross-coupled feedback maintains the previously established state.

The simulation therefore confirms that the implemented circuit performs the required three fundamental operations:

- **Set:**  $S = 1, R = 0 \Rightarrow Q = 1, QB = 0$
- **Reset:**  $S = 0, R = 1 \Rightarrow Q = 0, QB = 1$
- **Hold:**  $S = 0, R = 0 \Rightarrow$  previous state is retained

The verified SR latch will subsequently be used as the digital storage element at the output of the dynamic slicer. Therefore, its role in the complete circuit is to retain and provide the final logic decision generated by the analog comparison and regeneration stages.

### 3 Dynamic Slicer Operation

The implemented dynamic slicer operates as a clocked decision-making circuit. Its operation can be understood as a sequence of four fundamental stages:

$$\boxed{\text{Precharge} \rightarrow \text{Differential Comparison} \rightarrow \text{Regeneration} \rightarrow \text{Logic Latching}} \quad (4)$$

The complete schematic is shown in Fig. 3. The circuit can be divided into three main functional blocks: the differential input stage formed by  $M_1$  and  $M_2$  together with the clocked tail transistor  $M_0$ , the regenerative stage formed by  $M_5$ – $M_{11}$ , and the NOR-based SR latch at the output.

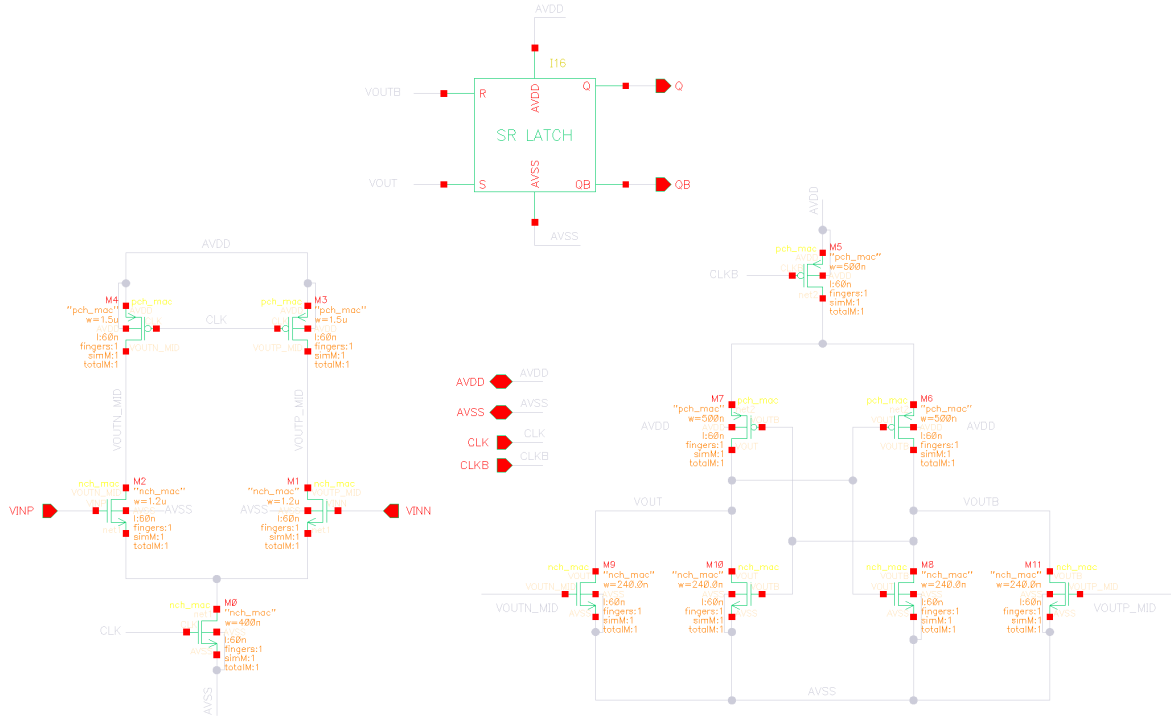


Figure 3: Complete schematic of the implemented dynamic slicer.

### 3.1 Precharge Phase: $CLK = 0$

During the precharge phase, the clock signals satisfy

$$CLK = 0, \quad CLKB = 1. \quad (5)$$

Consequently, the tail transistor  $M_0$  is turned OFF, while the PMOS transistors  $M_3$  and  $M_4$  are turned ON. The two dynamic internal nodes are therefore charged toward the supply voltage:

$$VOUTP\_MID \approx VOUTN\_MID \approx AVDD. \quad (6)$$

The equalization of these two nodes removes the voltage difference produced during the previous evaluation cycle and establishes a well-defined initial condition for the next comparison.

The precharge of the internal nodes also activates the NMOS transistors  $M_9$  and  $M_{11}$ , since their gate voltages are connected to  $VOUTN\_MID$  and  $VOUTP\_MID$ , respectively. As both gate voltages rise toward  $AVDD$ , both transistors turn ON and provide discharge paths from the output nodes to  $AVSS$ :

$$M_9 : \quad VOUT \rightarrow AVSS \quad (7)$$

$$M_{11} : \quad VOUTB \rightarrow AVSS. \quad (8)$$

Therefore, the two output nodes are initialized close to ground:

$$\boxed{VOUT \approx VOUTB \approx AVSS.} \quad (9)$$

At the same time,  $CLKB = 1$  causes the PMOS transistor  $M_5$  to turn OFF. Hence, the upper supply path of the regenerative stage is disconnected from  $AVDD$ , preventing the cross-coupled regenerative network from remaining active during the reset interval.

The precharge phase therefore establishes the following initial conditions:

$$\boxed{VOUTP\_MID \approx VOUTN\_MID \approx AVDD} \quad (10)$$

and

$$\boxed{VOUT \approx VOUTB \approx AVSS.} \quad (11)$$

Thus, before the beginning of the evaluation phase, both the dynamic input nodes and the output nodes are initialized to known states, ensuring that the next decision is determined by the newly applied differential input rather than by the previous comparison.



### 3.2 Evaluation and Differential Comparison

When the clock transitions to

$$CLK = 1, \quad CLKB = 0, \quad (12)$$

the circuit enters the evaluation (sampling) phase. The tail transistor  $M_0$  turns ON, while the precharge transistors  $M_3$  and  $M_4$  turn OFF. At the same time,  $M_5$  turns ON and reconnects the regenerative stage to  $AVDD$ .

The differential input pair  $M_1$  and  $M_2$  now compares  $V_{INP}$  and  $V_{INN}$ . Since the two transistors share the same tail current, a difference between their gate voltages produces a difference between their drain currents. Consequently, the two internal nodes no longer discharge at the same rate.

For example, when

$$V_{INP} > V_{INN}, \quad (13)$$

$M_2$  conducts more current than  $M_1$ . The corresponding internal node therefore discharges faster, creating a voltage difference between  $V_{OUTP\_MID}$  and  $V_{OUTN\_MID}$ :

$$V_{OUTP\_MID} \neq V_{OUTN\_MID}. \quad (14)$$

The important point is that the differential input does not directly produce a full logic-level output. Instead, it first creates a relatively small voltage imbalance between the two dynamic nodes. This imbalance is then used to initiate the regenerative process.

### 3.3 Regenerative Amplification

After the differential input pair has been enabled during the evaluation phase, the difference between  $V_{INP}$  and  $V_{INN}$  produces an unequal discharge of the two dynamic nodes,  $V_{OUTP\_MID}$  and  $V_{OUTN\_MID}$ . This voltage difference is then translated into an initial imbalance between the output nodes through  $M_9$  and  $M_{11}$ .

For example, consider the case

$$V_{INP} > V_{INN}. \quad (15)$$

Since  $M_2$  is driven by  $V_{INP}$  and  $M_1$  is driven by  $V_{INN}$ ,  $M_2$  conducts more strongly than  $M_1$ . Consequently,  $V_{OUTN\_MID}$ , which is connected to the drain of  $M_2$ , discharges faster than  $V_{OUTP\_MID}$ . Therefore,

$$V_{OUTN\_MID} < V_{OUTP\_MID}. \quad (16)$$

The two internal-node voltages control the NMOS transistors  $M_9$  and  $M_{11}$ , respectively. The gate of  $M_9$  is connected to  $V_{OUTN\_MID}$ , while the gate of  $M_{11}$  is connected to  $V_{OUTP\_MID}$ . As  $V_{OUTN\_MID}$  decreases,  $M_9$  becomes weaker and pulls  $V_{OUT}$  toward  $AVSS$  less strongly. In contrast,  $V_{OUTP\_MID}$  remains relatively higher, keeping  $M_{11}$  stronger and causing it to pull  $V_{OUTB}$  toward  $AVSS$ .

Consequently, an initial output imbalance develops in the direction

$$VOUT \uparrow, \quad VOUTB \downarrow. \quad (17)$$

This initial difference is then rapidly amplified by the regenerative network formed by the two cross-coupled CMOS inverters. The first inverter consists of  $M_7$  and  $M_{10}$ , while the second consists of  $M_6$  and  $M_8$ . Their input-output relationships are

$$VOUTB \rightarrow (M_7, M_{10}) \rightarrow VOUT \quad (18)$$

and

$$VOUT \rightarrow (M_6, M_8) \rightarrow VOUTB. \quad (19)$$

Because the two inverters are cross-coupled, any initial difference between  $VOUT$  and  $VOUTB$  is reinforced by positive feedback. As  $VOUTB$  decreases, the PMOS transistor  $M_7$  turns ON more strongly while the NMOS transistor  $M_{10}$  turns OFF, causing  $VOUT$  to rise further. The increase in  $VOUT$  simultaneously turns  $M_8$  ON and  $M_6$  OFF, which drives  $VOUTB$  even lower.

Therefore, the regenerative process can be represented as

$$VOUTB \downarrow \rightarrow VOUT \uparrow \rightarrow VOUTB \downarrow \rightarrow \dots \quad (20)$$

until the two outputs are driven toward complementary full-swing levels:

$$VOUT \approx AVDD, \quad VOUTB \approx AVSS. \quad (21)$$

The same mechanism occurs in the opposite direction when  $V_{INP} < V_{INN}$ . In that case, the roles of  $VOUTP\_MID$  and  $VOUTN\_MID$  are reversed, causing  $VOUT$  and  $VOUTB$  to regenerate toward the opposite logic states.

The regenerative stage therefore performs two essential functions:  $M_9$  and  $M_{11}$  convert the small voltage difference generated by the differential pair into an initial output imbalance, while the cross-coupled CMOS inverters formed by  $M_6$ – $M_8$  and  $M_{10}$  amplify this imbalance into a full-swing logic decision.

### 3.4 Logic Latching

After regeneration, the complementary signals  $VOUT$  and  $VOUTB$  are applied to the inputs of the NOR-based SR latch. The latch converts the regenerated result into the complementary logic outputs  $Q$  and  $QB$  and maintains the resulting state.

Thus, the complete decision path can be summarized as

$$V_{INP}, V_{INN} \rightarrow (VOUTP\_MID, VOUTN\_MID) \rightarrow (VOUT, VOUTB) \rightarrow (Q, QB). \quad (22)$$

The role of each stage can therefore be summarized as follows:

Table 2: Functional role of the main stages of the dynamic slicer.

| Circuit block           | Main function                                         |
|-------------------------|-------------------------------------------------------|
| $M_1, M_2, M_0$         | Differential comparison                               |
| $M_3, M_4$              | Precharge of the dynamic nodes                        |
| $M_5$                   | Clock-controlled supply for regeneration              |
| $M_9, M_{11}$           | Transfer of the differential imbalance to the outputs |
| $M_6, M_7, M_8, M_{10}$ | Positive-feedback regeneration                        |
| NOR SR latch            | Storage of the final logic decision                   |

The dynamic nature of the circuit is therefore essential to its operation: the circuit is first initialized during the precharge phase, then the differential input creates a small imbalance during evaluation, and finally the regenerative feedback rapidly converts that imbalance into a stable logic decision.

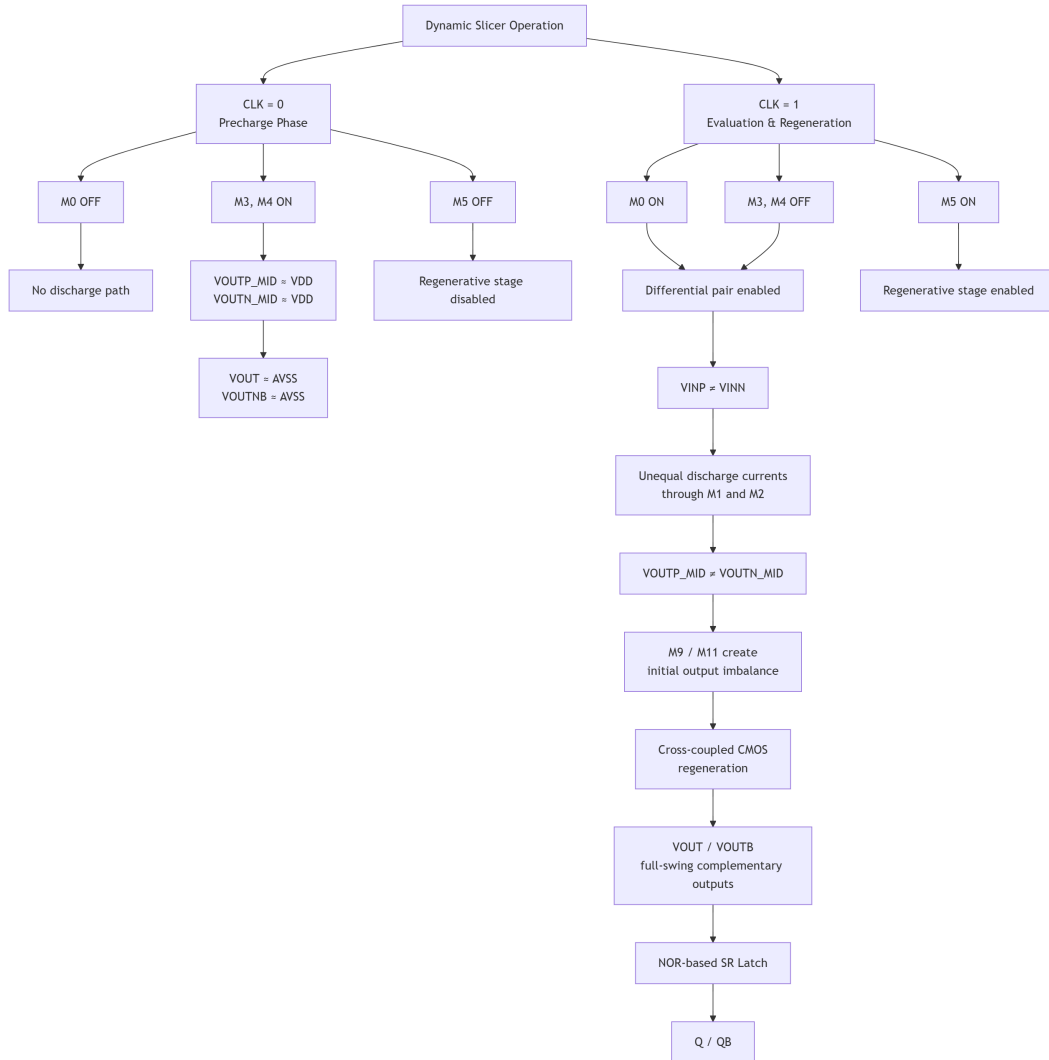


Figure 4: Functional flow of the dynamic slicer operation.

## 4 Part 1: Basic Dynamic Comparator Simulation and Characterization

### 4.1 Testbench Setup

The complete dynamic slicer was simulated using the testbench shown in Fig. 5. The supply voltage was set to

$$VDD = 1 \text{ V}, \quad (23)$$

while the common-mode input voltage was chosen as

$$VCM = \frac{VDD}{2} = 0.5 \text{ V}. \quad (24)$$

The differential input voltage was initially set to

$$VDIFF = 10 \text{ mV}. \quad (25)$$

Hence, the two differential input voltages are centered around the common-mode level:

$$V_{INP} = VCM + \frac{VDIFF}{2}, \quad (26)$$

$$V_{INN} = VCM - \frac{VDIFF}{2}. \quad (27)$$

For the initial simulation, this gives

$$V_{INP} = 505 \text{ mV}, \quad V_{INN} = 495 \text{ mV}. \quad (28)$$

The clock period was set to

$$PER = 1 \text{ ns}, \quad (29)$$

with complementary clock signals  $CLK$  and  $CLKB$ . The clock rise and fall times were set to  $0.1PER$ , the delay to  $0.5PER$ , and the pulse width to  $0.4PER$ , as specified in the lab instructions. The outputs were loaded with  $100 \text{ fF}$  capacitors to represent the capacitive load seen at the slicer outputs.

The transient analysis was performed for

$$t_{stop} = 2PER = 2 \text{ ns}. \quad (30)$$

For the subsequent parametric sweeps, the transient simulation time was maintained at  $2PER$  for every sweep point.

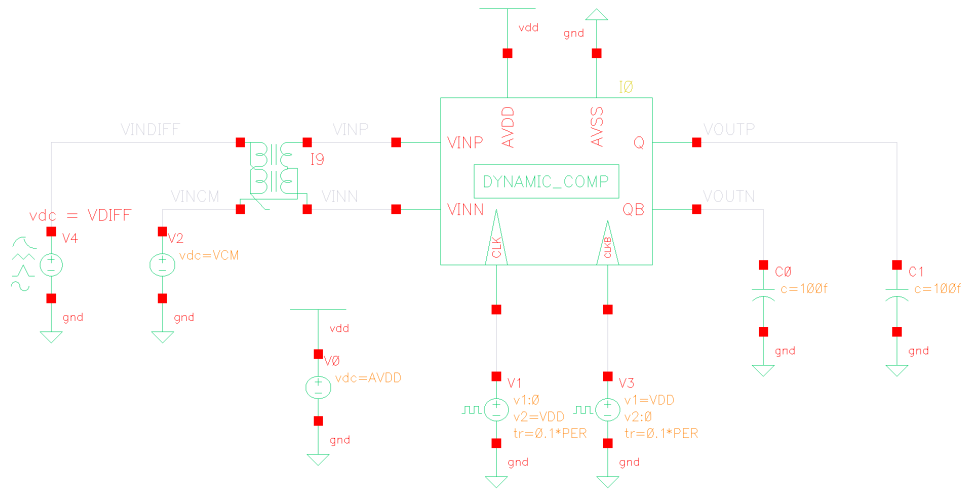


Figure 5: Testbench used for the basic dynamic slicer simulation.

## 4.2 Transient Simulation Results

The transient response of the clock, internal dynamic nodes, regenerative outputs, and final SR-latch outputs is shown in Fig. 6.

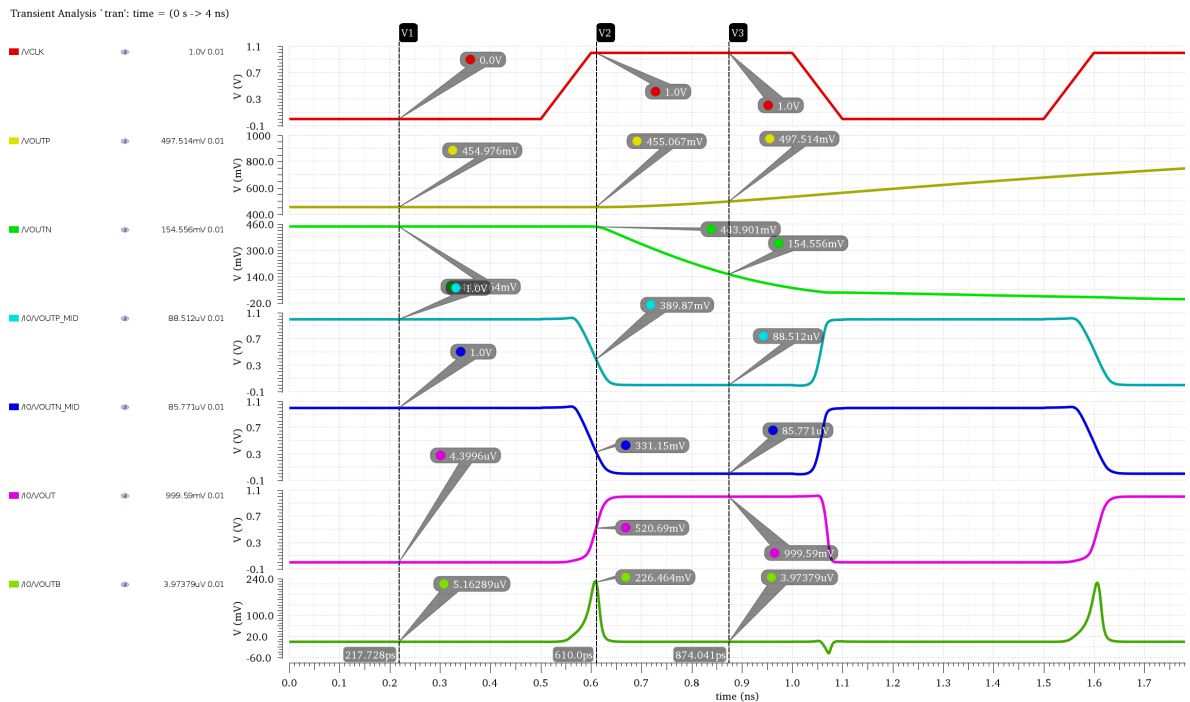


Figure 6: Transient waveforms of the dynamic slicer for  $PER = 1$  ns and  $VDIFF = 10$  mV.

The waveforms demonstrate the two-phase operation of the dynamic slicer. During the low-clock interval, the dynamic comparator is reset and the internal nodes are precharged. When  $CLK$  transitions high, the differential pair becomes active and the applied input difference produces unequal discharge rates at  $VOUTP\_MID$  and  $VOUTN\_MID$ . This initial imbalance is transferred to the regenerative output nodes, where the cross-coupled structure rapidly drives  $VOUT$  and  $VOUTB$  toward complementary logic levels.

An important observation is that the final outputs of the SR latch,  $VOUTP$  and  $VOUTN$ , do not necessarily reach their final rail voltages at the same rate as the internal regenerative nodes. The 100 fF load capacitances introduce additional charging and discharging time at these outputs. Therefore, the regenerative core can establish the logic decision before the loaded SR-latch outputs have completely settled to  $AVDD$  and  $AVSS$ .

During the following low-clock interval,  $M_5$  is turned OFF and the regenerative stage is disconnected from  $AVDD$ . At the same time, the internal comparator is reinitialized. Since  $VOUT$  and  $VOUTB$  are both driven low during this interval, the SR-latch inputs satisfy

$$S = R = 0. \quad (31)$$

Consequently, the NOR-based SR latch enters its hold state and preserves the decision obtained during the preceding evaluation phase. The SR-latch outputs can nevertheless continue moving toward their final voltage levels because the latch itself remains powered and its cross-coupled positive feedback continues reinforcing the stored logic state.

$$Q \uparrow \rightarrow VDD \quad (32)$$

$$Q_B \downarrow \rightarrow 0 \quad (33)$$

So you observe:

$$VOUTP \uparrow, \quad VOUTN \downarrow. \quad (34)$$

### 4.3 Delay Measurement

The decision delay was measured using the following calculator expression in Cadence Virtuoso:

```
delay(?wf1 VT("/VCLK") ?value1 0.5 ?edge1 "rising" ?nth1 1 ?td1 0
      ?wf2 abs((VT("/VOUTP") - VT("/VOUTN"))) ?value2 0.5
      ?edge2 "either" ?nth2 1 ?td2 nil ?stop nil ?multiple nil)
```

The expression measures the time interval between the first rising edge of  $VCLK$  at a voltage level of  $0.5V$  and the first occurrence at which the differential output magnitude satisfies

$$|VOUTP - VOUTN| = 0.5V. \quad (35)$$

Since  $VDD = 1V$ , the  $0.5V$  threshold corresponds to half of the supply voltage and is used as the decision criterion specified for the delay measurement. The resulting delay therefore represents the time required by the slicer, starting from the rising clock edge, to develop a sufficiently large differential output voltage to establish the logic decision.

The measured delay was:

$$t_{delay} = 502.5 \text{ ps}. \quad (36)$$

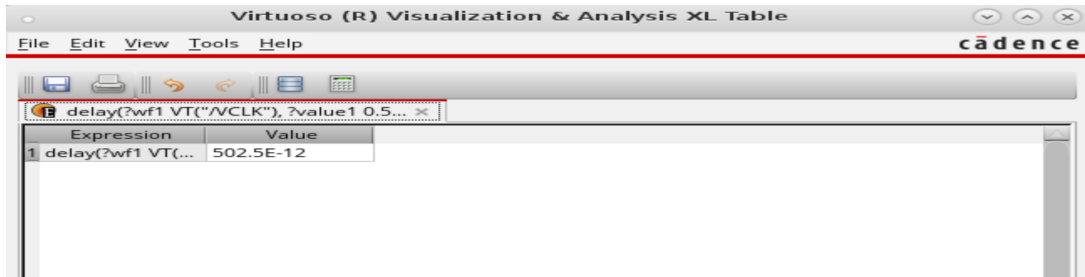


Figure 7: Measured delay of the dynamic slicer at  $VDIFF = 10 \text{ mV}$  and  $PER = 1 \text{ ns}$ .

The measured delay corresponds to approximately

$$\frac{t_{delay}}{PER} = \frac{502.5}{1000} \approx 50.25\% \quad (37)$$

of the clock period. Thus, the slicer reaches the specified  $0.5V$  decision criterion approximately halfway through the  $1 \text{ ns}$  clock period.

#### 4.4 Delay Dependence on Differential Input Voltage

To investigate the effect of the input differential magnitude,  $VDIFF$  was swept from 10 mV to 100 mV in 10 mV steps, as required by the lab. The resulting delay characteristic is shown in Fig. 8.

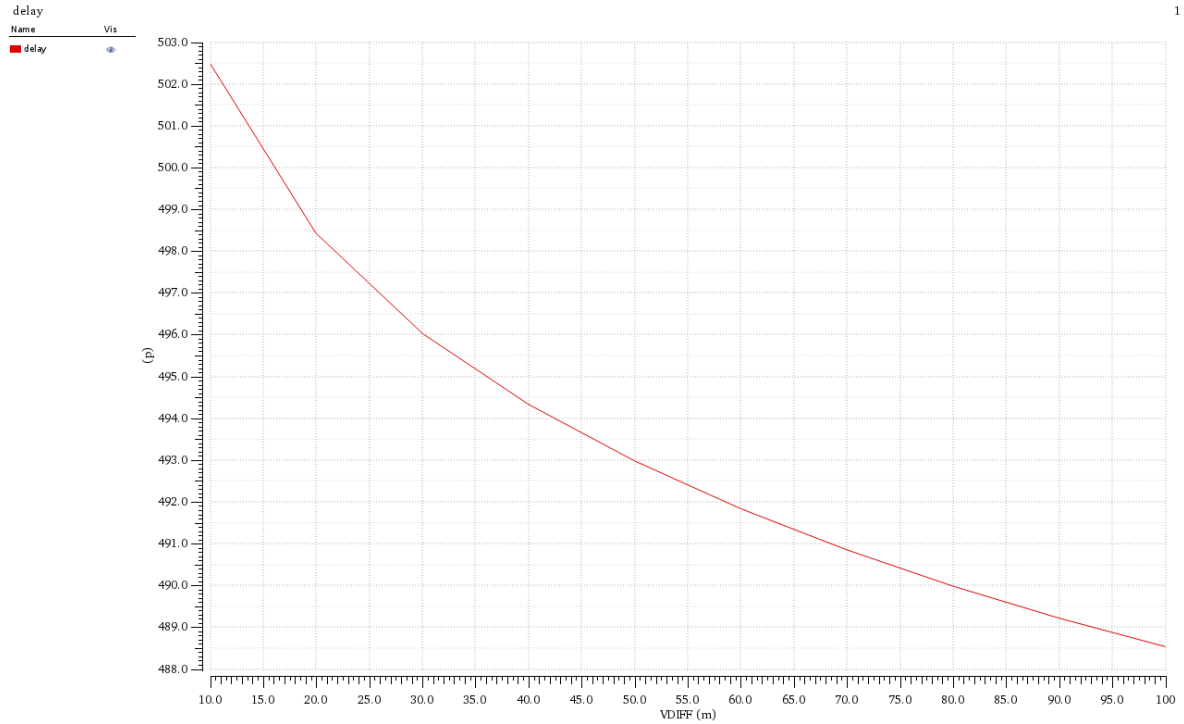


Figure 8: Dynamic slicer delay versus differential input voltage.

The delay decreases as  $VDIFF$  increases. The measured delay decreases from approximately

$$502.5 \text{ ps} \quad (38)$$

at

$$VDIFF = 10 \text{ mV}$$

to approximately

$$488.5 \text{ ps} \quad (39)$$

at

$$VDIFF = 100 \text{ mV}.$$



This behavior results from the differential pair generating a larger current imbalance when a larger input difference is applied. Consequently, the difference between  $VOUTP\_MID$  and  $VOUTN\_MID$  develops more rapidly, providing a larger initial imbalance for the regenerative stage. The cross-coupled regeneration can therefore reach the specified output decision threshold in a shorter time.

Although the delay decreases with increasing  $VDIFF$ , the overall change is relatively small over the investigated range. This indicates that the regenerative stage dominates the decision process once a sufficient initial imbalance has been established.

#### 4.5 Power Consumption

The instantaneous power consumption of the slicer was evaluated from the supply current according to

$$P(t) = VDD I_{DD}(t), \quad (40)$$

and the average power was obtained using the average supply current, as specified in the lab.

| Point              | Test               | Output         | Nominal                                                                               |
|--------------------|--------------------|----------------|---------------------------------------------------------------------------------------|
| Parameters: PER=1n |                    |                |                                                                                       |
| 1                  | LAB_04:TestBench:1 | VCLK           |  |
| 1                  | LAB_04:TestBench:1 | IT("/IO/AVDD") |  |
| 1                  | LAB_04:TestBench:1 | It             |  |
| 1                  | LAB_04:TestBench:1 | It_avg         | 31.52u                                                                                |
| 1                  | LAB_04:TestBench:1 | Pavg           | 31.52u                                                                                |

Figure 9: Power consumption of the dynamic slicer during transient operation.

The power waveform exhibits transient current peaks associated with the switching activity of the dynamic slicer. These peaks occur because the internal and output capacitances must be charged and discharged during the precharge, evaluation, and regeneration phases.

A capacitive current can be expressed approximately as

$$I_C = C \frac{dV}{dt}. \quad (41)$$

Therefore, faster voltage transitions produce larger instantaneous charging and discharging currents. These transient current peaks contribute directly to the average power consumption.

## 4.6 Average Power versus Clock Period

The clock period was then swept from 1 ns to 5 ns in 1 ns steps. For each sweep point, the transient simulation was performed over  $2PER$ , maintaining the same number of clock periods while changing the operating frequency.

The resulting average power is shown in Fig. 10.

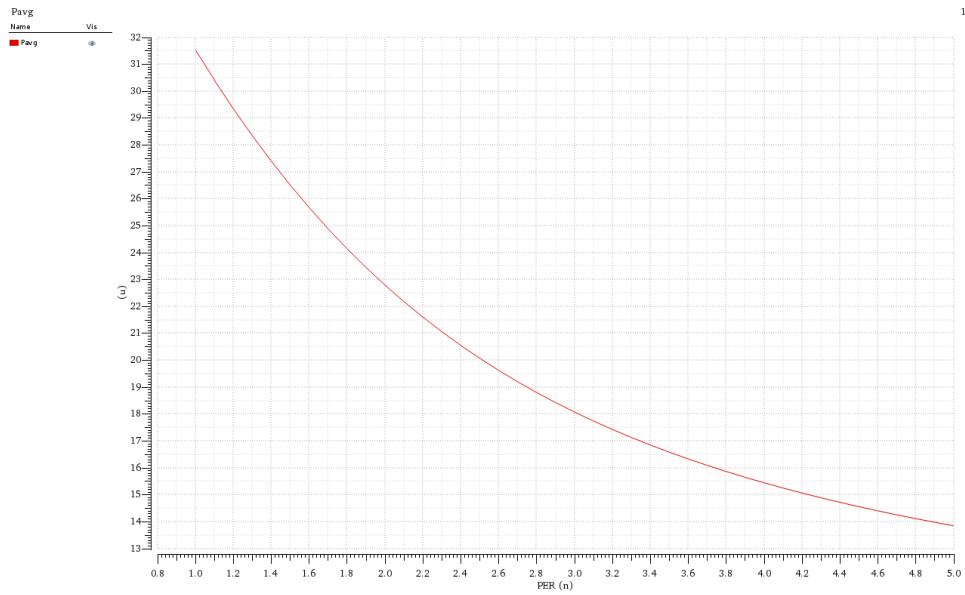


Figure 10: Average power consumption versus clock period.

The average power decreases as the clock period increases. The measured average power is approximately

$$P_{avg} \approx 31.52 \mu\text{W} \quad (42)$$

at

$$PER = 1 \text{ ns},$$

while it decreases to approximately

$$P_{avg} \approx 13.8 \mu\text{W} \quad (43)$$

at

$$PER = 5 \text{ ns}.$$

This trend is related to the reduction in switching frequency:

$$f_{CLK} = \frac{1}{PER}. \quad (44)$$

Thus, increasing  $PER$  reduces the frequency at which the dynamic slicer performs its charging, discharging, and regeneration operations. As a result, the energy dissipated by the switching activity is distributed over a longer time interval, reducing the average power.

The instantaneous supply-current waveforms provide further insight into this behavior.

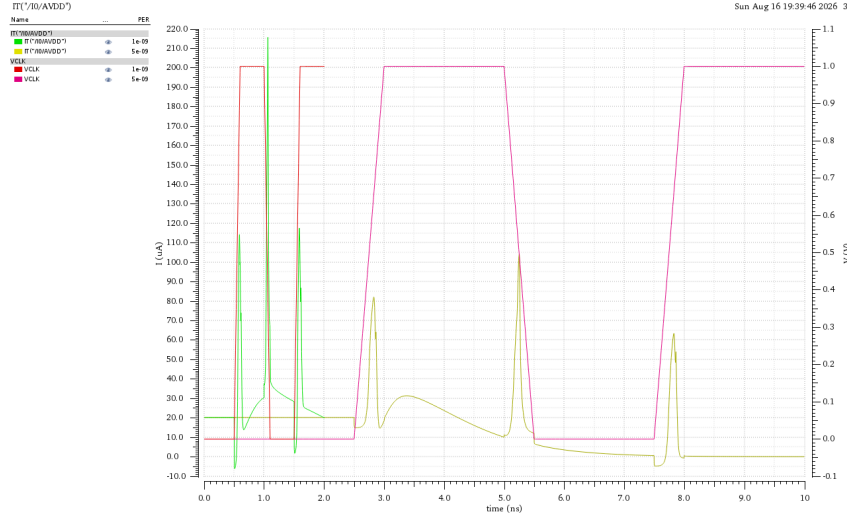


Figure 11: Supply-current response and clock waveform for  $PER = 1$  ns and  $PER = 5$  ns.

For the 1 ns case, the circuit operates at a higher frequency and must complete its charging, discharging, and regeneration transitions within a shorter time interval. Consequently, the voltage transitions are faster, resulting in larger transient current peaks according to

$$I_C = C \frac{dV}{dt}. \quad (45)$$

For the 5 ns case, the circuit operates more slowly and the voltage transitions occur over a longer interval. The corresponding transient current peaks are therefore smaller. In addition to the lower instantaneous current peaks, the switching frequency is also lower, which leads to the reduced average power observed in Fig. 10.

Overall, the results demonstrate the fundamental speed–power trade-off of the dynamic slicer: faster clock operation requires more rapid charging, discharging, and regeneration, producing larger transient currents and higher average power, whereas slower operation reduces the switching activity and average power.

## 5 Part 2: Overdrive (Overload) Recovery Test

The overdrive recovery test evaluates the ability of the dynamic slicer to correctly resolve a small input difference immediately after being subjected to a large differential input. This test is important because a strong previous decision can leave the dynamic and regenerative nodes far from their neutral state, potentially slowing the response to a subsequent small input.

### 5.1 Test Procedure

The testbench was configured such that a large positive differential input equal to  $V_{DD}$  is applied for three clock cycles. This strong input establishes a well-defined previous decision and provides the initial overdrive condition. In the following cycle, the differential input is reversed and reduced to a small negative value:

$$V_{DIFF} = -V_{SENS}. \quad (46)$$

The sensitivity parameter was swept according to

$$V_{SENS} = 1 \text{ mV} : 1 \text{ mV} : 10 \text{ mV}, \quad (47)$$

while maintaining

$$PER = 1 \text{ ns}, \quad V_{DD} = 1 \text{ V}, \quad T_{STOP} = 4PER. \quad (48)$$

The input waveform therefore consists of three cycles of a large positive differential input followed by a small opposite differential input in the fourth cycle. The purpose of this sequence is to determine whether the slicer can recover from the previous strongly regenerated state and correctly resolve the new, much smaller input.

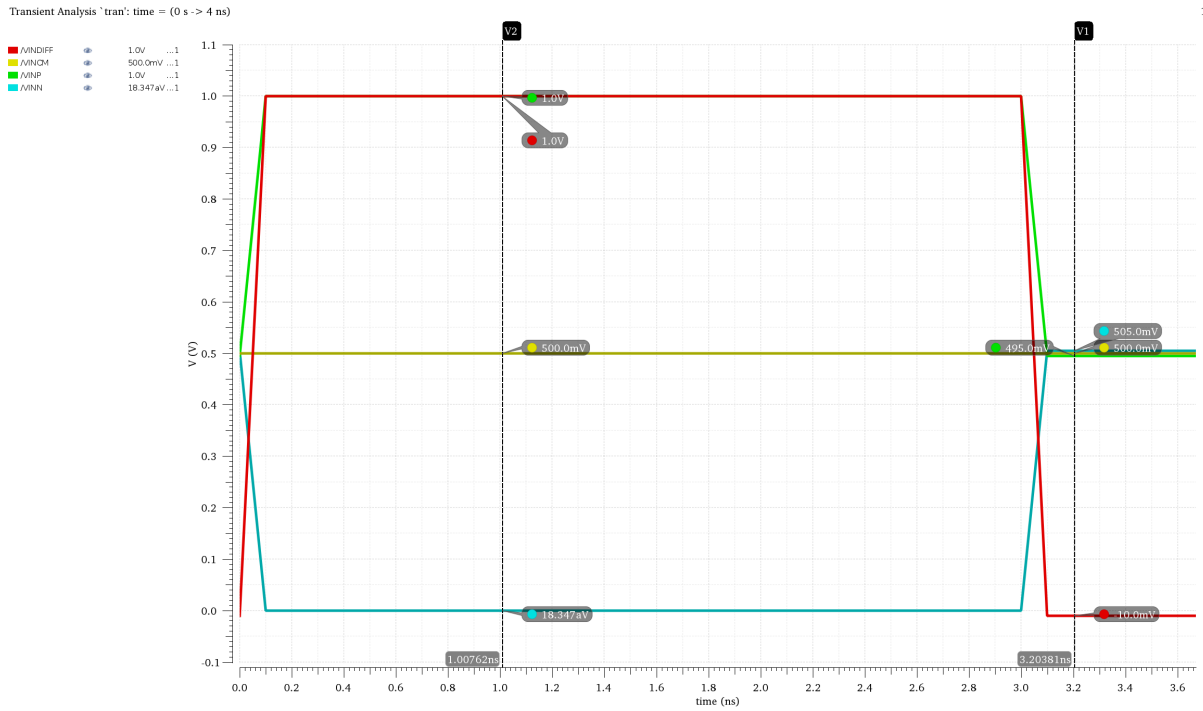


Figure 12: Input Waveforms

## 5.2 Sensitivity Measurement

The comparator sensitivity was evaluated using the following calculator expression provided in the lab:

```
value(VT("/IO/VOUT")-VT("/IO/VOUTB") 4*VAR("PER"))
```

The expression evaluates the differential output voltage at the end of the fourth clock period. The sign of

$$VOUT - VOUTB \quad (49)$$

is used to determine whether the comparator has successfully reversed its previous decision and resolved the small opposite input.

If the output difference has the expected sign, the comparator has successfully recovered from the overdrive condition. If the sign remains that of the previous decision, the comparator has failed to recover for the applied  $V_{SENS}$ . The lab defines the sensitivity as the minimum input magnitude at which the comparator is able to resolve the input correctly.

### 5.3 Simulation Results

The results of the  $V_{SENS}$  sweep are summarized in Table 3.

Table 3: Overdrive recovery results for different values of  $V_{SENS}$ .

| $V_{SENS}$ | $(V_{OUT} - V_{OUTB})$ at 4PER | Recovery   |
|------------|--------------------------------|------------|
| 1 mV       | +999.9 mV                      | Failed     |
| 2 mV       | −999.4 mV                      | Successful |
| 3 mV       | −999.4 mV                      | Successful |
| 4 mV       | −999.4 mV                      | Successful |
| 5 mV       | −999.4 mV                      | Successful |
| 6 mV       | −999.4 mV                      | Successful |
| 7 mV       | −999.4 mV                      | Successful |
| 8 mV       | −999.4 mV                      | Successful |
| 9 mV       | −999.4 mV                      | Successful |
| 10 mV      | −999.4 mV                      | Successful |

The results show a clear transition between 1 mV and 2 mV. For  $V_{SENS} = 1$  mV, the output difference remains positive and close to +1 V, indicating that the comparator has retained the previous decision and failed to resolve the small opposite input. When  $V_{SENS} = 2$  mV, the output difference becomes negative and approximately equal to −1 V, indicating successful recovery and correct resolution of the new input.

Therefore, with the available 1 mV sweep resolution, the minimum tested input that can be correctly resolved after the overdrive condition is

$$\boxed{V_{SENS,\min} \approx 2 \text{ mV}}. \quad (50)$$

Equivalently, the transition between successful and unsuccessful recovery occurs within the range

$$1 \text{ mV} < V_{SENS,\text{threshold}} \leq 2 \text{ mV}. \quad (51)$$

The result demonstrates that the dynamic slicer is capable of recovering from a strong previous decision and resolving an opposite differential input as small as approximately 2 mV under the applied test conditions.

## 6 Part 3: Offset Voltage Simulation

The final part of the lab investigates the input-referred offset voltage of the dynamic slicer. Ideally, a comparator should switch its output when the differential input voltage is zero,

$$V_{DIFF} = V_{INP} - V_{INN} = 0. \quad (52)$$

However, practical comparators exhibit an input-referred offset voltage,  $V_{OS}$ , which shifts the decision boundary away from zero. The offset can result from device mismatch and other circuit asymmetries. In a practical design, Monte Carlo mismatch simulation is commonly used to characterize the statistical distribution of the offset. In this lab, however, an artificial offset is intentionally introduced using a DC voltage source so that the accuracy of the offset-measurement method can be verified.

### 6.1 Testbench Setup

The offset-voltage testbench is shown in Fig. 13. A known DC offset of

$$V_{OS,applied} = 15 \text{ mV} \quad (53)$$

is intentionally inserted in the input path of the dynamic comparator. The differential input itself is generated as a slowly varying PWL signal that sweeps from

$$-V_{DIFF_{PEAK}} \quad (54)$$

to

$$+V_{DIFF_{PEAK}}, \quad (55)$$

with

$$V_{DIFF_{PEAK}} = 20 \text{ mV}. \quad (56)$$

The common-mode voltage and supply remain consistent with the previous simulations:

$$V_{DD} = 1 \text{ V}, \quad V_{CM} = \frac{V_{DD}}{2} = 0.5 \text{ V}. \quad (57)$$

The simulation is performed for

$$TEND = NCYCLE \cdot PER, \quad (58)$$

where

$$NCYCLE = 100, \quad PER = 1 \text{ ns}, \quad (59)$$

giving

$$TEND = 100 \text{ ns}. \quad (60)$$

The lab notes that increasing the number of cycles improves the measurement accuracy at the cost of increased simulation time.

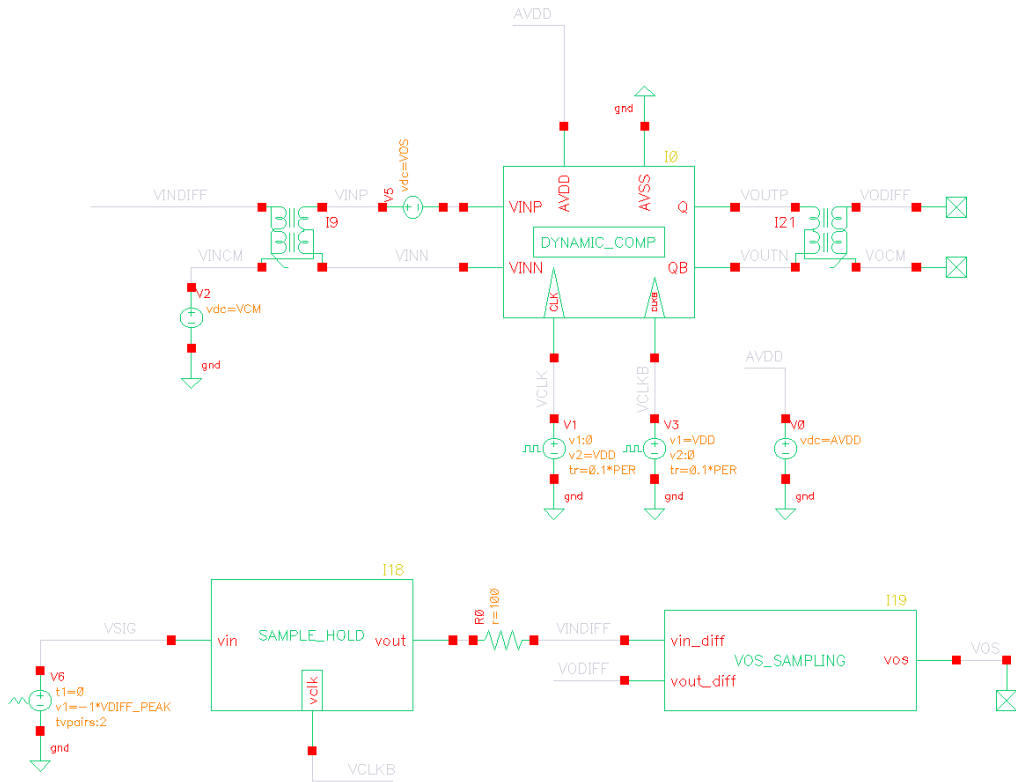


Figure 13: Testbench used for the dynamic slicer offset-voltage measurement.

## 6.2 Principle of Offset Measurement

The main idea of this experiment is to determine the differential input voltage at which the comparator changes its decision. If the comparator were ideal, the transition would occur at

$$V_{DIFF} = 0. \quad (61)$$



The presence of an input-referred offset shifts this transition point. Therefore, by slowly sweeping  $V_{DIFF}$  through the decision region and recording the input value corresponding to the comparator transition, the offset voltage can be inferred.

Conceptually, the measurement follows the sequence

PWL input  $\rightarrow$  comparator decision  $\rightarrow$  identify decision point  $\rightarrow$  infer  $V_{OS}$ .  
(62)

Because the dynamic slicer is a clocked circuit, the input cannot simply be read at an arbitrary time. The measurement must account for the fact that the comparator samples the input and requires a finite time to regenerate its output. For this reason, the lab provides two Verilog-A models to assist in the measurement.

### 6.3 Role of the Verilog-A Models

The first model, `sah_ideal`, implements an ideal sample-and-hold operation. It detects the specified clock transition and samples the input voltage at that instant, after which the sampled value is held at its output. This allows the input value associated with a sampling event to be preserved for subsequent processing.

The second model, `vos_sampling`, is used to determine the input-referred offset. It receives the differential input and the differential comparator output. The model delays the input signal by  $TCLK/4$  and monitors the comparator output until it crosses  $VDD/2$ . At that point, it samples the delayed differential input and calculates the offset using the sign convention

$$V_{OS} = -V_{DIFF, \text{decision}}. \quad (63)$$

The input delay used in the supplied model is

$$\frac{TCLK}{4}. \quad (64)$$

For

$$TCLK = 1 \text{ ns}, \quad (65)$$

this corresponds to

$$\frac{TCLK}{4} = 250 \text{ ps}. \quad (66)$$

This 250 ps value is part of the timing alignment implemented inside the offset-measurement model; it is not the measured propagation delay of the dynamic slicer. The latter was measured separately in Part 1 as approximately 502.5 ps. The supplied Verilog-A implementation defines this delay explicitly.

The comparator decision is detected when the differential output crosses

$$\frac{V_{DD}}{2} = 0.5 \text{ V}. \quad (67)$$

At this event, the corresponding input differential voltage is used to infer the offset. The supplied Verilog-A model performs this operation through the expression

```
@ (cross(V(vout_diff) - VDD/2, +1))
```

followed by the offset calculation in the model.

## 6.4 Offset Measurement Procedure

The input differential voltage is swept slowly over the range

$$-20 \text{ mV} \leq V_{DIFF} \leq +20 \text{ mV}. \quad (68)$$

he comparator therefore encounters input differences on both sides of its decision boundary. In the presence of the intentionally applied offset, the transition does not occur exactly at  $V_{DIFF} = 0$ .

The measurement process can be summarized as follows:

1. A known offset of 15 mV is introduced at the comparator input.
2. The differential input is swept slowly from  $-20 \text{ mV}$  to  $+20 \text{ mV}$ .
3. The dynamic slicer samples and regenerates the input decision according to the clock.
4. The `vos_sampling` model detects the comparator output transition.
5. The corresponding differential input value is extracted and converted into the input-referred offset.

Thus, the offset is not obtained by directly reading the 15 mV DC source. Instead, the measurement determines the shifted decision boundary of the comparator and uses that boundary to estimate the input-referred offset.

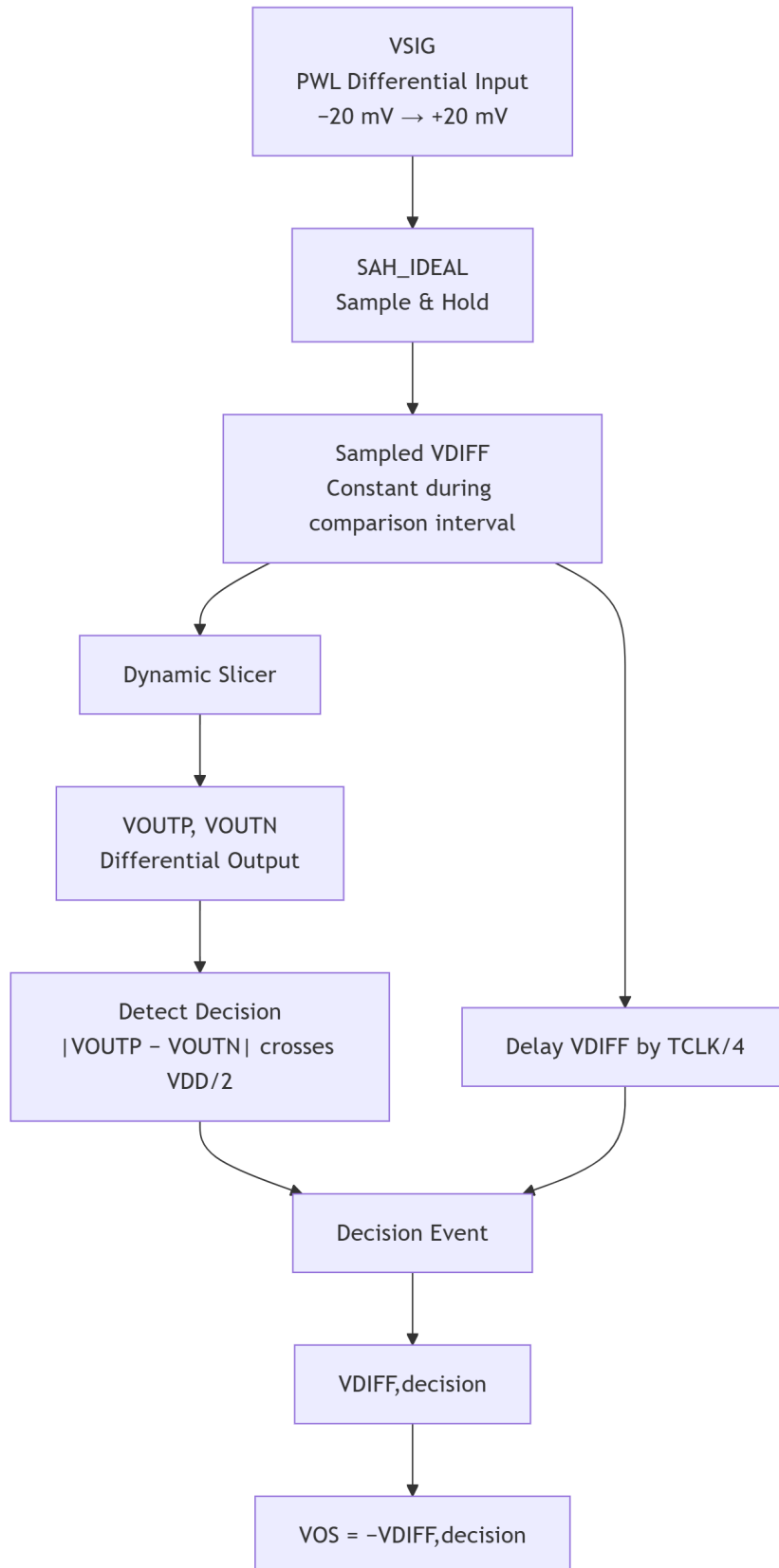


Figure 14: Flow of the input-referred offset voltage measurement.

## 6.5 Simulation Results

The resulting offset waveform and the corresponding differential-input behavior are shown in Fig. 15.

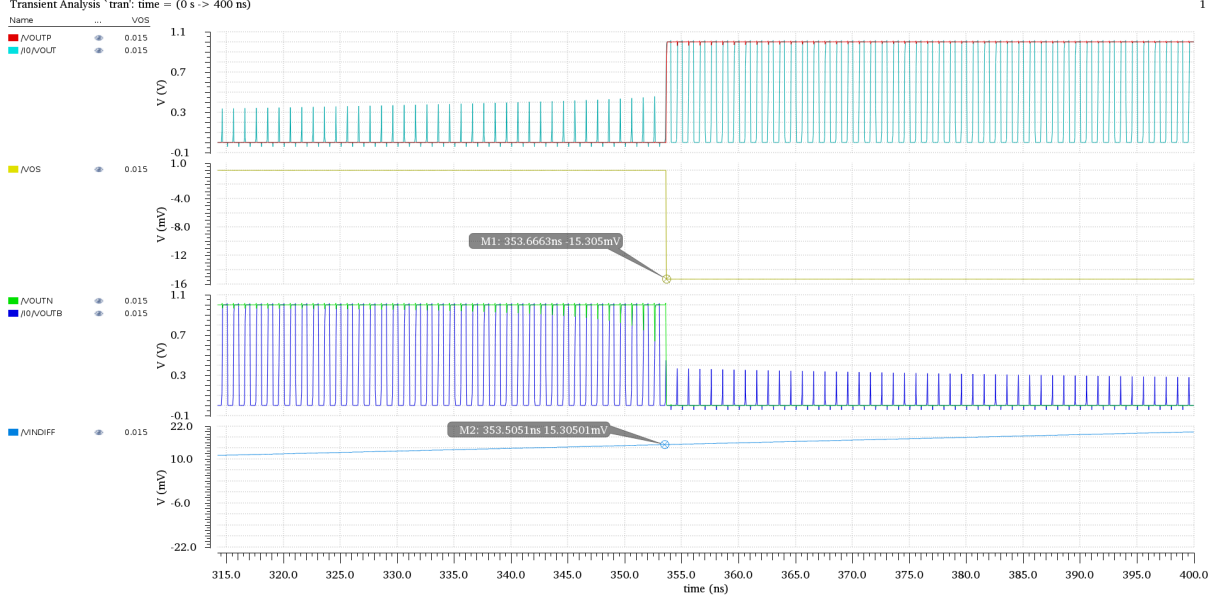


Figure 15: Offset-measurement result and differential-input sweep.

The measured input-referred offset obtained from the VOS output is

$$V_{OS, \text{measured}} = 15.31 \text{ mV}. \quad (69)$$

The intentionally applied offset was

$$V_{OS, \text{applied}} = 15.00 \text{ mV}. \quad (70)$$

Therefore, the absolute measurement error is

$$|V_{OS, \text{measured}} - V_{OS, \text{applied}}| = |15.31 - 15.00| = 0.31 \text{ mV}. \quad (71)$$

The corresponding percentage error is

$$\text{Error} = \frac{0.31}{15} \times 100 \approx 2.07\%. \quad (72)$$

The measured value is therefore in close agreement with the intentionally introduced 15 mV offset. This verifies that the sample-and-hold and offset-sampling methodology successfully extracts the input-referred offset of the dynamic slicer.